

Efficient Fiber-Chip Optical Coupling in Silicon Nitride Photonics Using Amorphous Silicon Grating Overlays

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Abstract We demonstrate α -Si/Si₃N₄ surface grating couplers with -1.6 dB experimental coupling loss. Inverse-designed multi-band couplers show -2 dB loss at O- and C-band wavelengths with -30 dB crosstalk. This enables compact and scalable fibre-chip interfaces for high-capacity optical communication and advanced photonic integrated circuits. ©2026 The Author(s)

INTRODUCTION

Surface grating couplers are recognized to be an appealing and scalable solution for light routing on and off photonic integrated circuits (PICs). Forming optical input/output (I/O) interfaces directly on a photonic chip surface facilitates device prototyping and rapid testing using automated wafer-level accessories. Surface gratings also offer extra freedom for dense multi-port interconnections [1], advanced integration using optical interposers [2], and far-field beam control [3].

The silicon nitride (Si₃N₄) platform is an excellent choice for the realization of complex PICs due to ultra-low losses, wide transparency, and mature deposition processes and fabrication, which is now available through open-access foundries. However, fibre-chip optical coupling using surface gratings remains challenging for Si₃N₄ photonics. The lower core-to-cladding refractive index contrast hinders efficient light extraction from the optical waveguide, requiring long structures. In particular, Si₃N₄ gratings have significantly reduced strength (compared to silicon (Si) counterparts), resulting in low radiation efficiency and poor overlap between the near-Gaussian fibre mode and out-radiated grating beam [4,5].

To date, single-layer full- and shallow-etch surface gratings remain an attractive choice due to their fabrication simplicity and ease-to-scale towards different platform stacks, wavebands, or polarizations [6-8]. However, they provide only a limited degree of design freedom, which inherently limits coupling performance to -4 dB to -6 dB for uniform and up to -2.5 dB for apodized couplers [5,7]. To access sub-dB loss, which are required by many applications, advanced design strategies coupled with additional fabrication steps need to be adopted. Recently, profound design strategies to improve optical coupling performance and enhanced functionalities include

utilization of backside reflectors [9], multi-etch waveguide profiles [10], multi-level grating arrangement [11], or top overlays [12-17]. The latter approach is particularly appealing in the recent years. The use of top overlays compensates for low grating strength through localized material deposition, enhancing radiation efficiency while maintaining low-loss light propagation. Moreover, this approach facilitates simple front-end integration of Si-compatible materials into complex photonic systems without introducing processing overhead.

In this work, we report on the design and fabrication of amorphous silicon (α -Si) assisted Si₃N₄ surface grating couplers with close to -1 dB loss for a single-band operation and -2.1 dB for a multi-band operation.

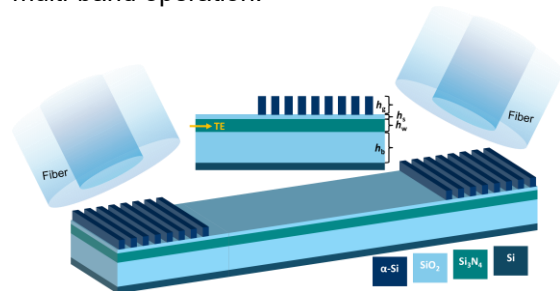


Fig. 1. Three-dimensional (3D) schematics of surface grating couplers realized on hybrid α -Si/Si₃N₄ photonic platform.

Hybrid surface grating couplers

The 3D schematic of α -Si/Si₃N₄ surface gratings is shown in Fig. 1. The devices are realized on a Si₃N₄ platform having a waveguide thickness, h_w , on top of buried oxide (BOX) layer, h_b . The grating is fully defined in the α -Si layer, h_g , separated from a light-guiding nitride layer by oxide buffer, h_s . Single- and multi-band couplers are developed for operation at original (O-band, 1.31 μ m) and conventional (C-band, 1.55 μ m) optical windows, typically harnessed by short-reach datacom services, passive optical networks, and long-

distance telecom systems.

Uniform α -Si surface gratings

Surface grating couplers face efficiency limits from poor directionality and mode mismatch. Apodized designs improve the performance but typically require tight critical dimensions and/or complex fabrication. On the other hand, uniform gratings offer robustness yet suffer from limited overlap due to the exponential near-field decay. To address this problem, we developed a hybrid surface grating coupler that leverages inter-layer mode interference between vertically separated Si_3N_4 bus waveguide and a full-etch α -Si grating [13]. The grating functionality is well-described by a virtual directional coupler design to engineer constructive interference and shape the out-radiated field. This approach overcomes the exponential decay of uniform gratings, achieving a near-Gaussian profile with field overlap exceeding the typical $\sim 80\%$ limit. Unlike conventional apodized designs, it preserves a uniform geometry, avoids sub-120 nm features and multi-step fabrication, and offers enhanced robustness and compatibility with scalable foundry manufacturing processes.

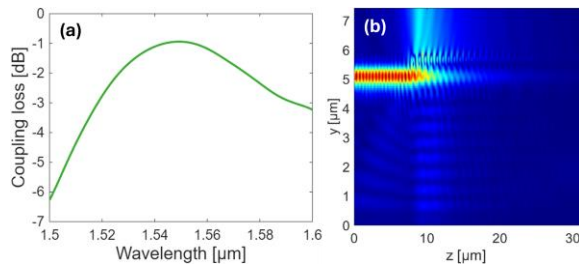


Fig. 2. (a) Coupling loss as a function of the wavelength for uniform inverse-designed α -Si/ Si_3N_4 coupler. Inset. Electric field distribution along the light diffraction axis.

A proof-of-concept design elaborated through the rigorous 3D FDTD simulations, showed near-unity grating-to-fibre overlap and a coupling loss of about -1.3 dB at 950 nm to favour integration with III/V quantum dot sources. Moreover, inverse designed grating couplers optimized via metaheuristic algorithms showed coupling loss as low as -0.97 dB in the C-band wavelengths [14], as shown in Fig. 2a. This confirms the scalability and design robustness of the proposed methodology across different wavelengths. Figure 2b shows electric field distribution of the uniform surface grating coupler implemented in a hybrid α -Si/ Si_3N_4 photonic platform.

Beam-focalized metamaterial couplers

Hybrid material platforms enable additional design freedom to optimize the performance of surface grating couplers. Besides the uniform waveguide couplers, we present a novel grating coupler architecture, where the input light from the

Si_3N_4 waveguide is evanescently coupled into the top fully etched α -Si surface grating [17]. The high refractive index contrast of the α -Si layer breaks the vertical symmetry of the structure, significantly enhancing upward radiation ($>90\%$) and suppressing substrate leakage ($<10\%$), thereby improving grating directionality. The uniform grating coupler is optimized through systematic variation of the grating period and duty cycle using 2D and 3D simulations in Ansys Lumerical. The design targets efficient coupling to a standard SMF-28 fibre (MFD = $10.4 \mu\text{m}$) under TE polarization. Simulated coupling losses reach -2.4 dB at $1.55 \mu\text{m}$ and -1.7 dB at $1.31 \mu\text{m}$, respectively. Fabricated devices are characterized via back-to-back transmission measurements, yielding a peak coupling loss of -3.56 dB near a reference wavelength of 1550 nm . A scanning electron microscope image (SEM) of the fabricated uniform device is presented in Fig. 3, together with its enlarged view as inset.

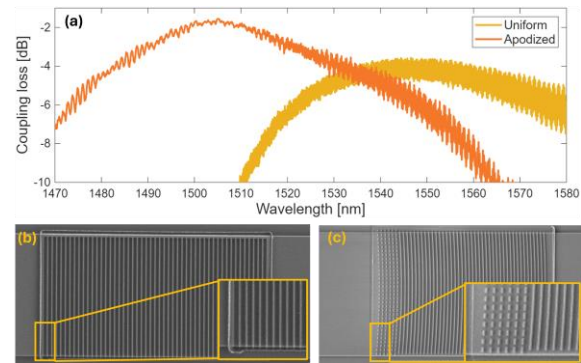


Fig. 3. (a) Measured coupling loss as a function of the wavelength for uniform and apodized α -Si/ Si_3N_4 grating couplers operated at C-band wavelengths. Scanning electron microscope images of fabricated surface grating couplers: (b) uniform and (c) apodized devices. Insets: enlarged views of fabricated grating structures.

The performance of the uniform surface grating couplers is further improved by integrating sub-wavelength grating (SWG) metamaterials with beam focalization concept [17]. In this design scheme, α -Si SWG structures provide extra simultaneous control over the amplitude and phase of the out-radiated field, enabling accurate engineering of the near-field profile for quasi-optimal overlap with the Gaussian-like mode of a standard optical fibre. A self-imaging focusing mechanism is additionally employed to concentrate the radiated beam into an SMF-28 fibre located above the chip surface. This combined strategy overcomes key limitations of Si_3N_4 grating couplers, including weak radiation strength and poor directionality, thereby improving overall performance of Si_3N_4 -based optical interfaces. By imposing a converging phase front, the radiated beam is focalized to the required MFD at the fibre interface, enhancing overlap, while

compensating for reduced scattering strength near the grating input. The apodized focusing grating was optimized using an iterative 2D FDTD workflow driven by a genetic algorithm, followed by final dimensional refinement via rigorous 3D FDTD simulations. This optimization yields a coupling loss of -1.0 dB, corresponding to an improvement of 1.4 dB compared to the uniform grating designs. Experimentally, at a wavelength of 1507 nm, measured fibre-chip coupling loss is -1.6 dB, shown in Fig. 3a. The SEM of fabricated apodized beam-focalized coupler with SWG metamaterials in α -Si layer is shown in Fig. 3c, together with enlarged device SWG features

Multi-band grating couplers

The hybrid α -Si/Si₃N₄ platform introduces a powerful route toward spectrally engineered surface grating couplers, enabling simultaneous low-loss fibre-chip interfacing and multi-band functionality within a single device. By leveraging the high index contrast of the α -Si overlay combined with the broadband transparency of Si₃N₄ the platform enables precise control over the local scattering strength and phase response, which is essential for tailoring radiation profiles across multiple optical channels.

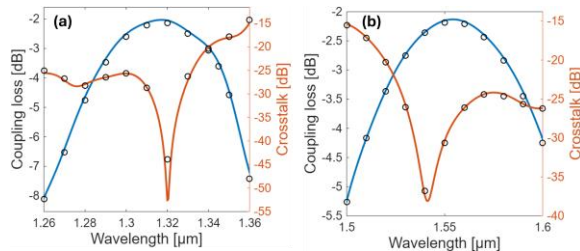


Fig. 4. Coupling loss and crosstalk as a function of the wavelength for dual-band surface grating couplers in α -Si/Si₃N₄ platform. (a) O-band and (b) C-band optical communication windows. Note: solid colored lines correspond to 2D FDTD designs, while black circles to 3D FDTD data.

We developed inverse-designed dual-band surface grating couplers targeting passive optical network (PON) wavelengths: 1.55 μ m and 1.31 μ m.

To efficiently explore complex high-dimensional and non-convex parameter landscape associated with hybrid multi-layer surface gratings, metaheuristic optimization algorithms are integrated within the inverse design framework [14]. The design problem is formulated as a multi-objective optimization, where coupling loss, spectral isolation (crosstalk), and bandwidth uniformity are simultaneously maximized. Evolutionary-based algorithms are employed to globally explore the parameter space and avoid local minima. Compared to conventional forward-designed grating couplers, the proposed approach enables the synthesis of complex radiation

profiles that maintain high mode overlap with the fibre Gaussian mode across multiple discrete wavelengths, while suppressing undesired spectral leakage.

For the dual-band device operating at 1.31 μ m and 1.55 μ m, the optimized structure achieves coupling losses of -2.1 dB at both wavelengths (Figs. 4a,b), with inter-channel crosstalk suppressed to $\sim$ -30 dB. The corresponding 1 dB bandwidths reach 64 nm and 49 nm for the 1.55 nm and 1.31 μ m channels, respectively, indicating robust spectral performance suitable for practical deployment. The achieved performance, combining near -2 dB coupling loss with low crosstalk and broadband optical response, establishes hybrid α -Si/Si₃N₄ inverse-designed gratings as a scalable solution for next-generation multi-band photonic interfaces in optical communications and beyond.

Conclusions

We reported recent progress in the development of low-loss surface grating couplers for Si₃N₄ PICs using top α -Si overlays. Optimized, uniform and apodized devices showed the potential for fibre-chip coupling losses approaching 1 dB, with experimental coupling loss of -1.6 dB. Moreover, by leveraging inverse-design approach, we developed multi-band couplers with losses of -2 dB and low crosstalk of -30 dB. These results pave the way for the development of scalable low-loss grating-coupled optical interfaces for Si₃N₄ integrated photonics, targeting applications as diverse as datacom, 10 Gbps PONs, and modern telecom systems.

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